

V5 scientific integrity and reviewer implementation audit

Evidence boundary

Frozen source commit: `d5fefaa0cf7a31cf7d29830ed5b070e1d2dac3a2`. Current physical arrays, held-event predictions and training labels were not altered. MIKE rainfall/terrain/reference arrays and the public LarNO checkpoint are upstream materials, clearly attributed; A/B/C coupled simulations, fitted residual estimators and their diagnostics are this project's calculations. The manuscript does not claim all data are original observations.

Verified findings

The raw rainfall sum and active-cell mean use different masks. Geometric slope inspection finds no zero/negative conduit slope and 626 values near the design floor. No outfall has its own coordinate record. Spatial maps mark connecting terminals only. Peak-map differences are not instantaneous residuals. The held-event network increment is 0.047773 mm, event SD 0.037333 mm and sampling-seed macro SD 0.001819 mm. All are computed, not borrowed from the LarNO paper.

Implementation status

Review item	Status	Evidence or limitation
Rainfall mask and volume	Recomputed	576 intervals; verified numerical redistribution, not surveyed roof mask
Peak-map meaning	Corrected	$\max(C) - \max(B)$ explicitly separated from $C(t) - B(t)$
Slope zeros and outfalls	Recomputed	0 nonpositive; 626 near floor; outfall coordinates absent
Event and seed uncertainty	Recomputed	40 event-seed gains; event bootstrap conditional on exchangeability
Inference perturbations	Reframed	OOD sensitivity separated from retrained feature benefit
Native exchange formula/sign	Unit check only	Prescribed-head test double; no routed pipe benchmark
Full joint internal-step water ledger	Pending	30-minute cumulative records and final report only
Broader paired events / weather independence	Pending	Nine readable cases unpaired; dates/families unverified
Multi-event rainfall treatment / network capacities / tailwater	Pending	Only existing single-event setting sensitivities completed
Coarse prior / shifted rain / cross-scenario retraining	Pending	No new predictive claims
Unclamped depth error changes and extreme LarNO tails	Partially addressed	LarNO full-range maps and extremes rebuilt; clamp counts retained; full raw-depth change diagnostics still need inference replay
Authors and redistribution rights	Author action	Do not invent author declarations or upstream permission

Exact numerical evidence

File	SHA256
extended_study/output/reviewer_major_revision_v5/diagnostics/event_seed_gain.csv	38a470bd1594e725bdce7cfa3799018a417986646b2ea70a983ac0e83a6397a0
extended_study/output/reviewer_major_revision_v5/diagnostics/hydrograph_diagnostics.csv	d650c81cee3961e409672859b3a2cac4e95139ed320ab504b7763b93a59cf2c8
extended_study/output/reviewer_major_revision_v5/diagnostics/larno_extremes.json	1593c5d449e64c7b9bec5fbad13e9310b1bd3a1a4eb1f1796bb2214d3e482428
extended_study/output/reviewer_major_revision_v5/diagnostics/map_display_audit.csv	8d9d7b3d1190d4f5b1727e4274f56af29a4800472f184104d406a709a2d80ebe
extended_study/output/reviewer_major_revision_v5/diagnostics/native_exchange_unit_check.json	ef1f6529a83bec2b1b036952b12c66d62c718779d178d379db5a21281eb73f5c
extended_study/output/reviewer_major_revision_v5/diagnostics/network_geometry.json	4f694b48565d4e5f3a045afdfc8650d5ffe4faa85cca9756dbbfde62e0f46f8
extended_study/output/reviewer_major_revision_v5/diagnostics/rainfall_event.csv	9fafcaf913163ff5342fe24622daec99a62dbba229939cce0b828921af6cec5
extended_study/output/reviewer_major_revision_v5/diagnostics/rainfall_steps.csv	927a98aa697bf07c0d02e5d4f9aaa008c4e20f2a4b00c5a616ff62eca01f00bf
extended_study/output/reviewer_major_revision_v5/diagnostics/signed_residual_diagnostics.csv	e17a7c27b821c13d73f4ac5cc8a288e1ef2f6fb38242c703cflc7d649ef9af50
extended_study/output/reviewer_major_revision_v5/diagnostics/summary.json	3953ac835be6b93713e6efc661dbcea8432a2f7794c452848a3b17b40fa6f711
extended_study/output/reviewer_major_revision_v5/diagnostics/water_ledger_saved_times.csv	f694e6605b502d64038af47257aea863c4637c10e8d388fed95dd9176598a6b6

Reproduction commands

```
python extended_study/audit_reviewer_v5_evidence.py
python extended_study/test_v5_native_exchange.py
python extended_study/generate_reviewer_v5_figures.py
python extended_study/generate_reviewer_v5_documents.py
python extended_study/render_reviewer_v5_documents.py
```

Primary training remains `extended_study/run_final_hybrid_major_revision.py` (fixed 140 iterations, no residual clipping). Historical V3 tuning is not a reproduction command for the V4/V5 primary results. Figures 1/2/6/7/9/10/11 and supplementary maps are rebuilt by the V5 figure script; unchanged physical-quality, hydrograph and peak-map illustrations originate from the frozen V4 package. Figure 5's data formula is audited in `plot_reviewer_v3_physics.py`. Full CSV evidence remains alongside concise manuscript tables; removing duplicate plots does not remove inconvenient results.

Writing sources and source verification

The revision follows concrete, evidence-led prose and distinguishes methods, results and interpretation. It does not manufacture an author voice, field observations or stronger conclusions to evade AI detection. Sources consulted: Manchester Academic Phrasebank (<https://www.phrasebank.manchester.ac.uk/>), KKKKhazix/human-writing (<https://github.com/KKKKhazix/human-writing>), petergyang/no-ai-slop (<https://github.com/petergyang/no-ai-slop>). These are editorial guidance, not scientific evidence.

MIKE's coupled origin is established in Section 4.1 of the supplied Cao et al. paper, DOI 10.1016/j.jhydrol.2026.135686. The current public model card documents 13 released input channels and the 64/16 Futian split (https://github.com/holmescao/LarNO/blob/main/huggingface/model_card.md). Current Itzi stable documentation resolves to 26.6, while the frozen calculations use 25.4; the current documentation version must not be represented as the executed engine. Normal-depth outfalls are conceptual hydraulic boundary conditions, not measurements of receiving-water levels.

Submission readiness

This revision improves presentation and evidence transparency. It does not close the pending physical validation and scenario-generalisation requests. Numerical consistency checks alone are not grounds to declare the study calibrated or ready for unconditional submission.

V6 editorial change record

The physical arrays and learned predictions are unchanged. Figure 2 is copied byte-for-byte; Figures 3-5 retain data and layouts with terminology-only label changes. Figure 6 retains its fields, time selection and colour bounds. The skill plot is removed because Table 2 already reports its results. The event scatter is replaced by a full supplementary inventory. Sensitivity time series are recomputed from the six existing paired simulations, and their temporal mean absolute residual and final-volume differences are checked against the original summary CSV (tolerances 0.001 mm and 1 m3). Figure labels use LarNO; implementation provenance remains in the supplement.

Terminology source: Fan et al. (2017), <https://doi.org/10.1155/2017/2819308>. The established term coupled 1D-2D hydrodynamic model motivates the consistent 1D/2D wording; it does not imply independent validation of this implementation. Event68 rationale describes measured characteristics retrospectively, not a fabricated prospective selection rule. Pending physical experiments remain pending.

V7 correction to prior quality claims

Numerical quality assessment

Global routing continuity errors range from -0.178% to -0.060%, and reported nonconverging steps from 0.00% to 0.04%. Continuity error is a numerical water-budget residual, not a disconnected-pipe count. A nonzero residual is not automatically a model failure, but small global values do not establish local accuracy. In event68 the reported node N01520 has a 63.01% inflow-normalised local imbalance. Its rounded cumulative inflow is about 114 m3, corresponding to approximately 71.8 m3 of imbalance; this is not 63% of whole-network water. The same node's inflow table reports 170.331% because that table uses outflow as its denominator. These are different normalisations of the same budget discrepancy, not conflicting results. The existing aggregate screening therefore does not close the numerical validation. The reported local errors must be checked against node-level exchanges and shorter-step results before the labels can be described as locally validated.

Event	Global balance (%)	Nonconverging steps (%)	Top local node	Local balance (% inflow)	Assessment
event1	-0.083	0.02	N00184	47.07	Local review required
event20	-0.178	0.0	N00184	69.6	Local review required
event65	-0.111	0.0	N00184	65.45	Local review required
event66	-0.101	0.0	N00184	58.75	Local review required
event67	-0.066	0.01	N01520	65.04	Local review required
event68	-0.06	0.04	N01520	63.01	Local review required
event69	-0.068	0.01	N01520	52.93	Local review required
event70	-0.063	0.01	N01520	56.86	Local review required

Interpretation and required follow-up

The EPA manual (Section 8.5) treats continuity as a numerical diagnostic, calls for investigation of excessive local errors at relevant nodes, and recommends time series at one-minute or shorter resolution for initial instability screening. Its example of a 10% excessive global error is not a universal safe threshold. The repository limits of 2% routing/nonconvergence and 0.5% combined error are project screens, not proof of physical validity. A printed 0.00% is rounded and does not prove zero failed iterations.

A nonconverging-step percentage measures steps that fail the iteration stopping criterion; it is neither a percent of wrong depths nor an estimate of mass loss. The current report prints two decimal places and cannot recover exact failed-step counts or their timing.

Next acceptance: instrument inflow, outflow, storage and signed exchange at the flagged nodes; repeat event68 and the event with the largest local discrepancy with reduced maximum routing steps (0.5, 0.25, 0.125 s) and reviewed head tolerance/iteration settings; compare node hydrographs, local budgets and surface labels. Do not alter the physical network merely to improve a diagnostic. No convergence rerun was executed by this audit, and no label or learned prediction has been replaced.

Sources

- EPA manual, Sections 8.5 and 9.1
- Engine 5.2.4 local ranking formula
- Engine 5.2.4 inflow-table formula